

Approaches for Hybrid Search

The integration of Graph DB and Vector DB is where the "hybrid" power truly shines. Here are common patterns:

1. Filter-then-Search (Structured Filtering + Unstructured Semantic Search):

- **Process:**

1. **Graph DB Query:** Start by querying the Graph DB to apply structured filters or traverse relationships to narrow down the entities (e.g., patients, diagnoses, medications). This returns a set of structured data or entity IDs.
2. **Vector DB Query:** For the identified entities, pass their associated unstructured text (e.g., clinical_notes of specific patients) to the Vector DB to perform a semantic search.
3. **Combine/Present:** Combine the structured information with the semantically relevant unstructured snippets.

- **Scenario Example:** Question 3: "Find patients who have both 'Type 2 Diabetes' and 'Hypertension' as initial diagnoses. Among these, identify anyone whose chief complaint in *any* clinical note semantically matches 'unusual thirst and frequent urination'."

- *Graph DB:* Find patients with "Type 2 Diabetes" AND "Hypertension."
- *Vector DB:* For *these specific patients*, semantically search their clinical_notes for "unusual thirst and frequent urination."

2. Search-then-Filter/Contextualize (Unstructured Semantic Search + Structured Contextualization):

- **Process:**

1. **Vector DB Query:** Begin by performing a semantic search on unstructured data (e.g., all clinical_notes) to find relevant snippets or associated entity IDs (e.g., patient_id from the note).
2. **Graph DB Query:** Use the identified entity IDs to query the Graph DB to retrieve structured attributes, related entities, or apply further structured filters (e.g., "only patients on a specific medication").
3. **Combine/Present:** Integrate the semantically retrieved notes with the structured context from the Graph DB.

- **Scenario Example:** Question 1: "Identify all patients who reported symptoms like 'dizziness' or 'fatigue' in their free-text clinical notes, and who are currently on 'Lisinopril'."
 - *Vector DB:* Find all notes containing "dizziness" or "fatigue" and get their associated patient_id.
 - *Graph DB:* For *these specific patients*, check if "Lisinopril" is in their medications_on_record and retrieve their initial_diagnoses.

3. Parallel Querying and Fusion:

- **Process:**
 1. **Parallel Queries:** Send parts of the user query to both the Graph DB and the Vector DB simultaneously.
 2. **Result Fusion:** Combine and rank the results from both databases using a sophisticated fusion strategy. This could involve re-ranking based on relevance scores from both systems, or joining on common identifiers (e.g., patient_id).
- **Scenario Example:** Question 5: "Suppose a patient (e.g., Richard Frank, P002) presents with a new complaint of 'sudden blurry vision'. Retrieve all their past clinical_notes where 'Type 2 Diabetes' or 'Hypertension' were reviewed. Additionally, conceptually, if we had an external knowledge base...".
 - *Graph DB:* Query for patient P002's initial_diagnoses and filter clinical_notes by date/diagnosis review.
 - *Vector DB:* Semantically search patient P002's clinical_notes (and potentially an external knowledge base) for "blurry vision" or related concepts.
 - *Fusion:* Present the historical context alongside semantically retrieved notes and external knowledge base suggestions.

4. Graph-Enhanced chunk metadata:

- **Process:** While not a real-time query approach, this is a crucial *offline* strategy. When creating embeddings for the clinical_notes for the Vector DB, incorporate structured context from the Graph DB. For example, instead of just embedding the note content, you could embed a combined string like: "Patient P001, Male, 81, Diagnoses: Hypertension. Medications: Lisinopril. Clinical Note: [note content here]." This enriches

the semantic meaning of the notes, allowing for more precise vector searches later.

By adopting these approaches, the application can leverage the strengths of both database types, providing a comprehensive and intelligent clinical decision support system.

How the Application Decides Which Database to Query First

The decision largely depends on the **nature of the user's query** and whether the primary information being sought is structured/relational or unstructured/semantic.

1. Analyzing the Query:

- **Natural Language Processing (NLP)**
 - Advanced applications use NLP techniques like Named Entity Recognition (NER), intent classification, and semantic parsing to understand the user's input.
 - **Structured Entities/Attributes:** If the query explicitly mentions structured attributes (e.g., "male patients," "age over 60," "diagnosis of Hypertension," "medication Lisinopril"), the system recognizes these as Graph DB-filterable criteria.
 - **Unstructured/Semantic Concepts:** If the query involves free-text descriptions or semantic meaning that needs to be extracted from unstructured content (e.g., "dizziness," "fatigue," "unusual thirst," "blurry vision" found in `clinical_notes`), the system flags these for Vector DB processing.
 - **Keywords and Patterns:** Simpler systems might rely on keyword spotting or predefined query patterns. For example, if "clinical notes" or "symptoms reported" are in the query, it might indicate a need for Vector DB interaction.

2. Ask the LLM

You can leverage a prompt to ask the LLM to guide you based on the user question what hybrid sequence to use. Give it a context of what your graph and vector db have.

3. Determining the Primary Information Source:

- **Querying Graph DB First (for Structured-Primary Queries):**
 - If the query starts with structured filters or requests relationships between clearly defined entities (e.g., "All *male patients* over 60 who have *Hypertension*..."), the Graph DB is the initial entry point.

It efficiently filters down the dataset to a relevant subset of patients based on these structured criteria.

- **Example (from previous questions):** "For all male patients over 60 years old who have an initial diagnosis of 'Hypertension', trace their clinical_notes chronologically..." Here, gender, age, and initial_diagnoses are structured, making the Graph DB the natural starting point.

- **Querying Vector DB First (for Unstructured-Primary Queries):**

- If the query's core involves understanding or matching a free-text snippet or semantic concept found primarily within unstructured data (e.g., "patients who reported 'fatigue' or 'nausea' in their *clinical notes*..."), the Vector DB is queried first. It performs a semantic similarity search to identify relevant notes or patient IDs.
- **Example (from previous questions):** "Identify all patients who reported symptoms like 'dizziness' or 'fatigue' in their free-text clinical notes, and who are currently on 'Lisinopril'." The symptom in notes is the primary search criterion.